

Status of the ESA Gaia mission

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Gaia reveals the origin and evolution of the Milky Way

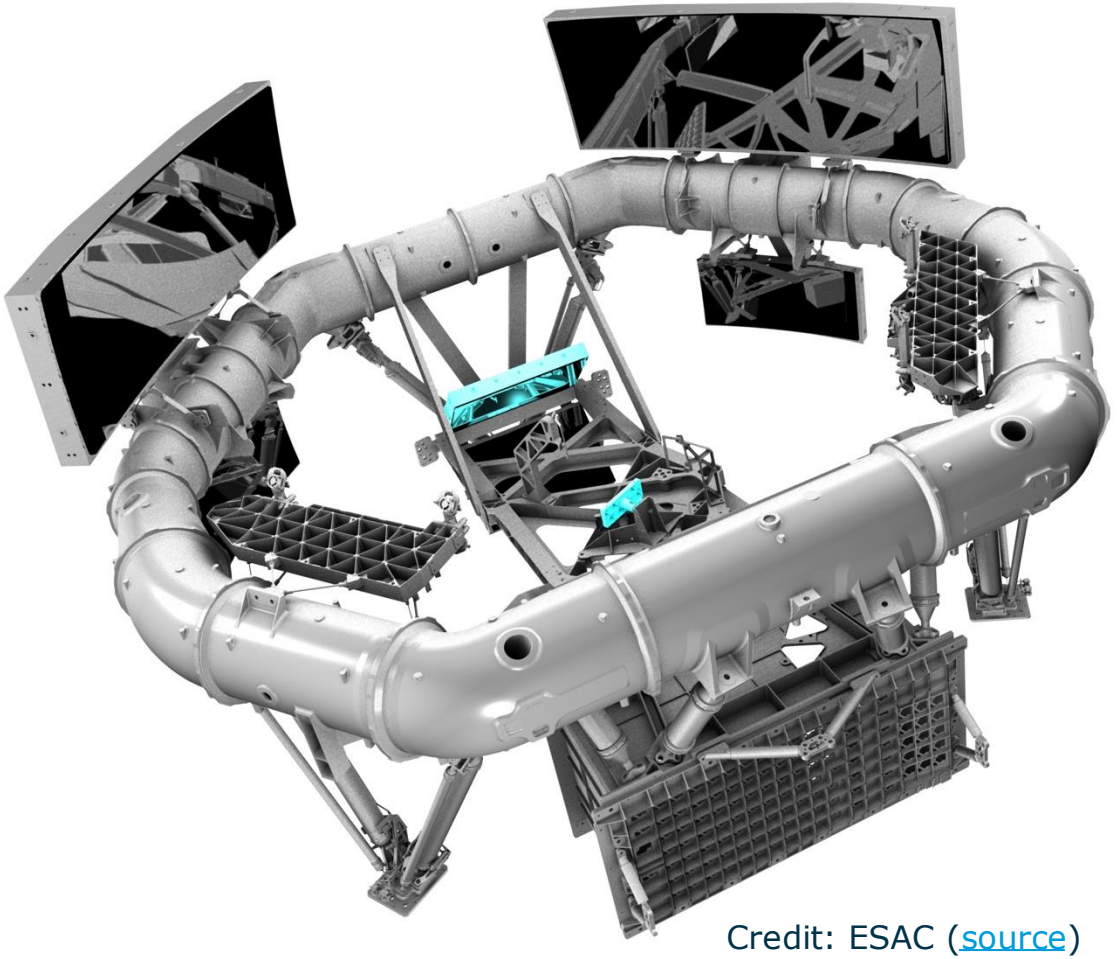
- Stellar positions and distances
  ➡ Galactic structure
- Stellar motions
  ➡ Galactic kinematics
- Stellar parameters, compositions, ages
  ➡ Galactic evolution
- Optical all-sky survey with 3 instruments:
  Astrometry, Photometry, Spectroscopy
- Gaia observes ~1% of the stars in the Galaxy
- Gaia survey enables rich complementary science



Credit: ESA/Gaia/DPAC, Stefan Payne-Wardenaar (source)

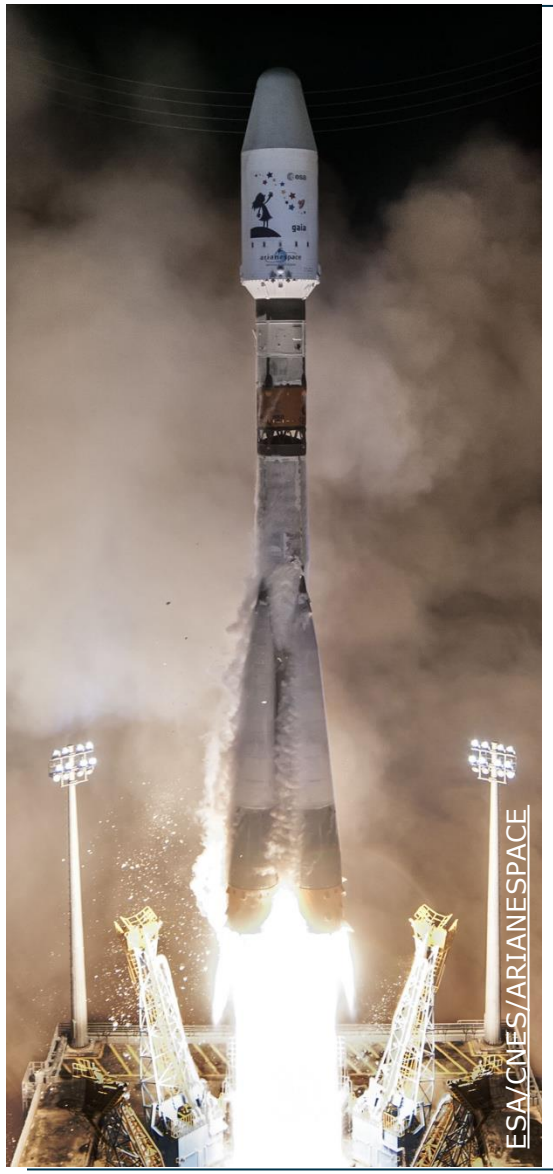
The Gaia spacecraft is 10 m wide and has 2 telescopes



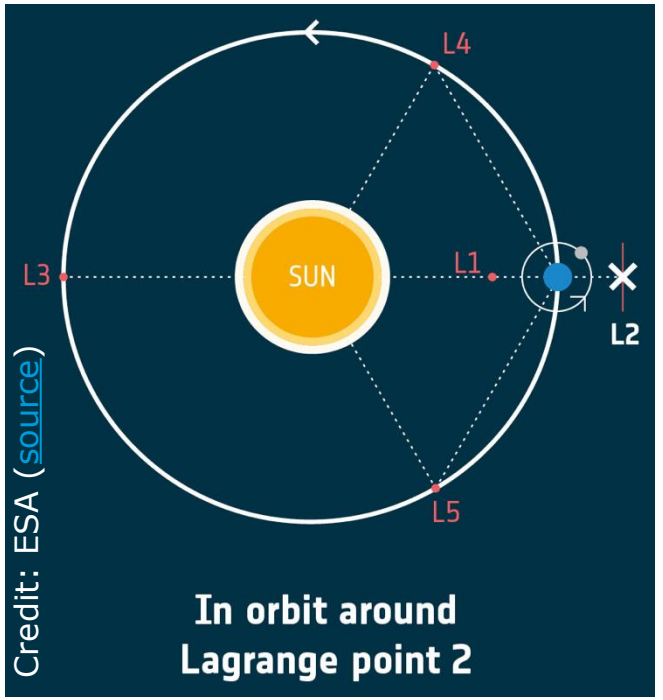


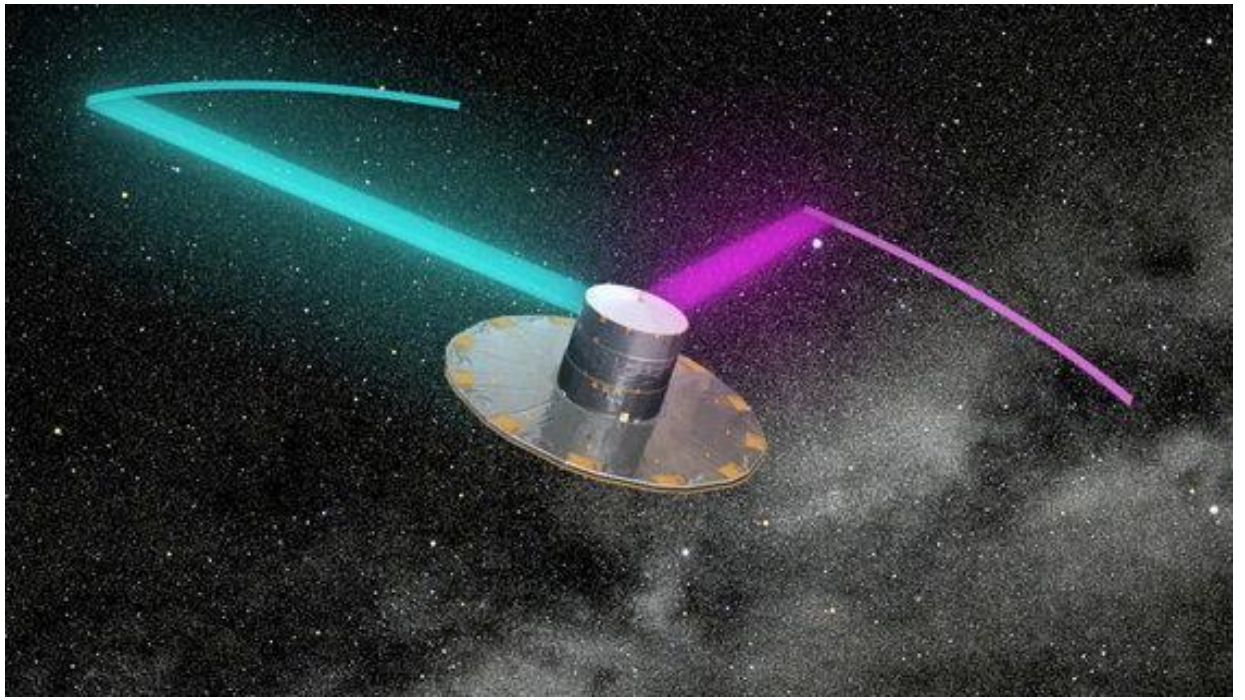
Credit: ESAC (source)

Gaia was launched to L2 in December 2013

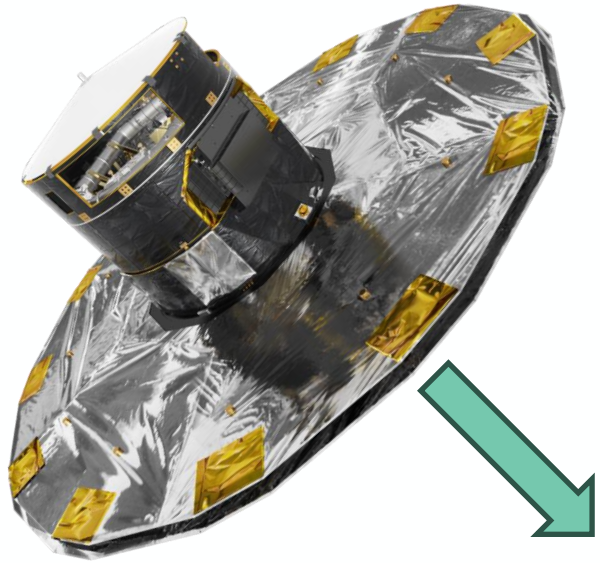


- 10.5 years of science observations from July 2014 to January 2025
- Gaia was constantly spinning and scanning the sky
- Surveying star-like sources, e.g. stars, galaxies, quasars, asteroids, …
- Magnitudes range G ≈ 3-21





Gaia data flow from the spacecraft to the ESA archive



Credit: ESA/ATG medialab (source)

Gaia consortium incl. ESA

Gaia DPAC



Credit: ESA (source)

Archive

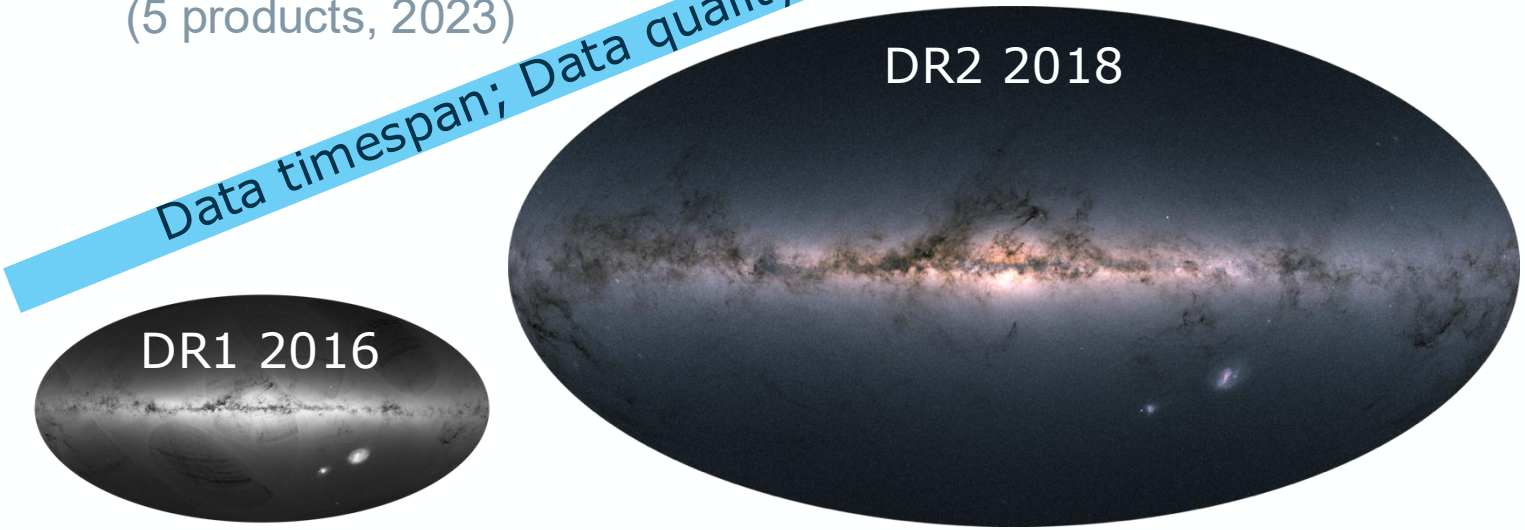
There were 3 intermediate Gaia data releases so far

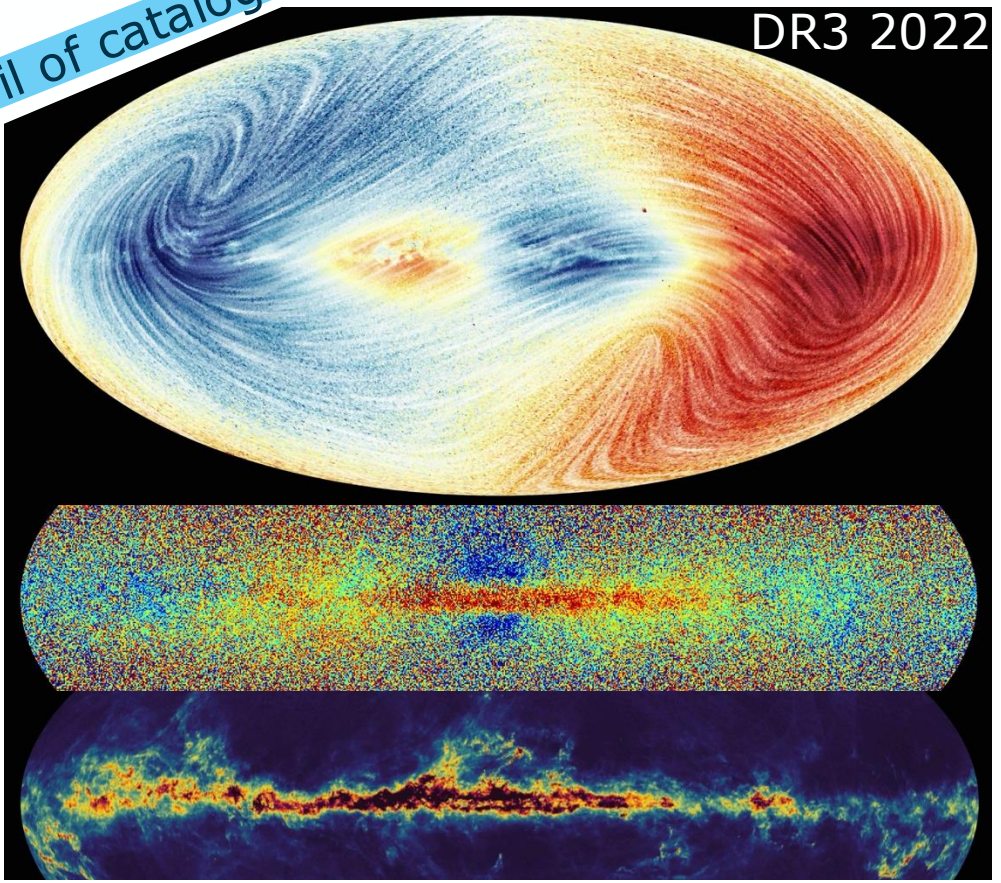
- Iterative process of incremental data releases
- There is no proprietary period on science data for the Gaia collaboration

- https://www.cosmos.esa.int/web/gaia/data

- Early DR3 (2020)
- Focused Product Release (5 products, 2023)

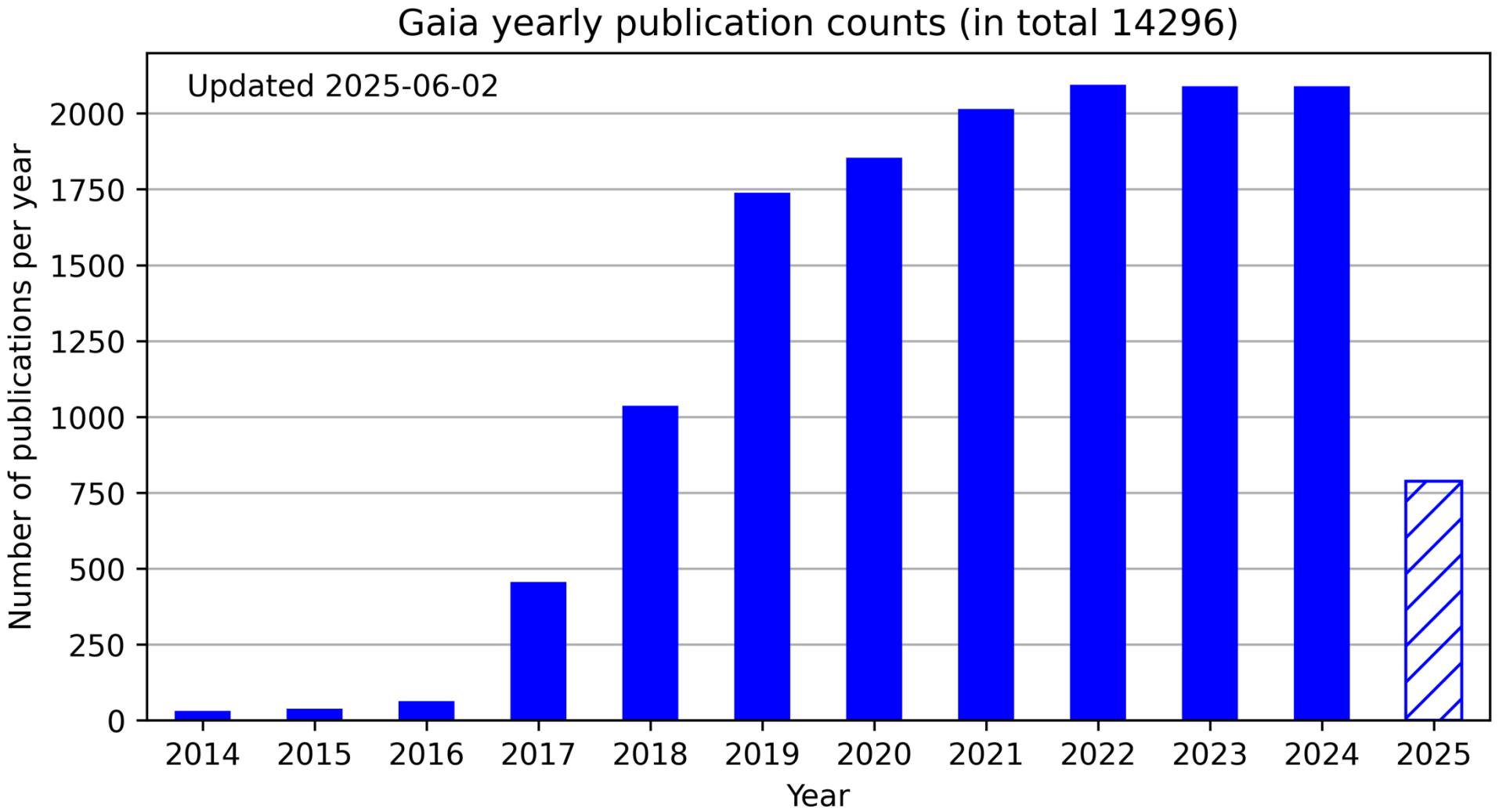
Data timespan; Data quality; Data volume; Number and detail of catalogues





source

Gaia is a very productive science mission



Credit: ESA/Gaia/DPAC (source)

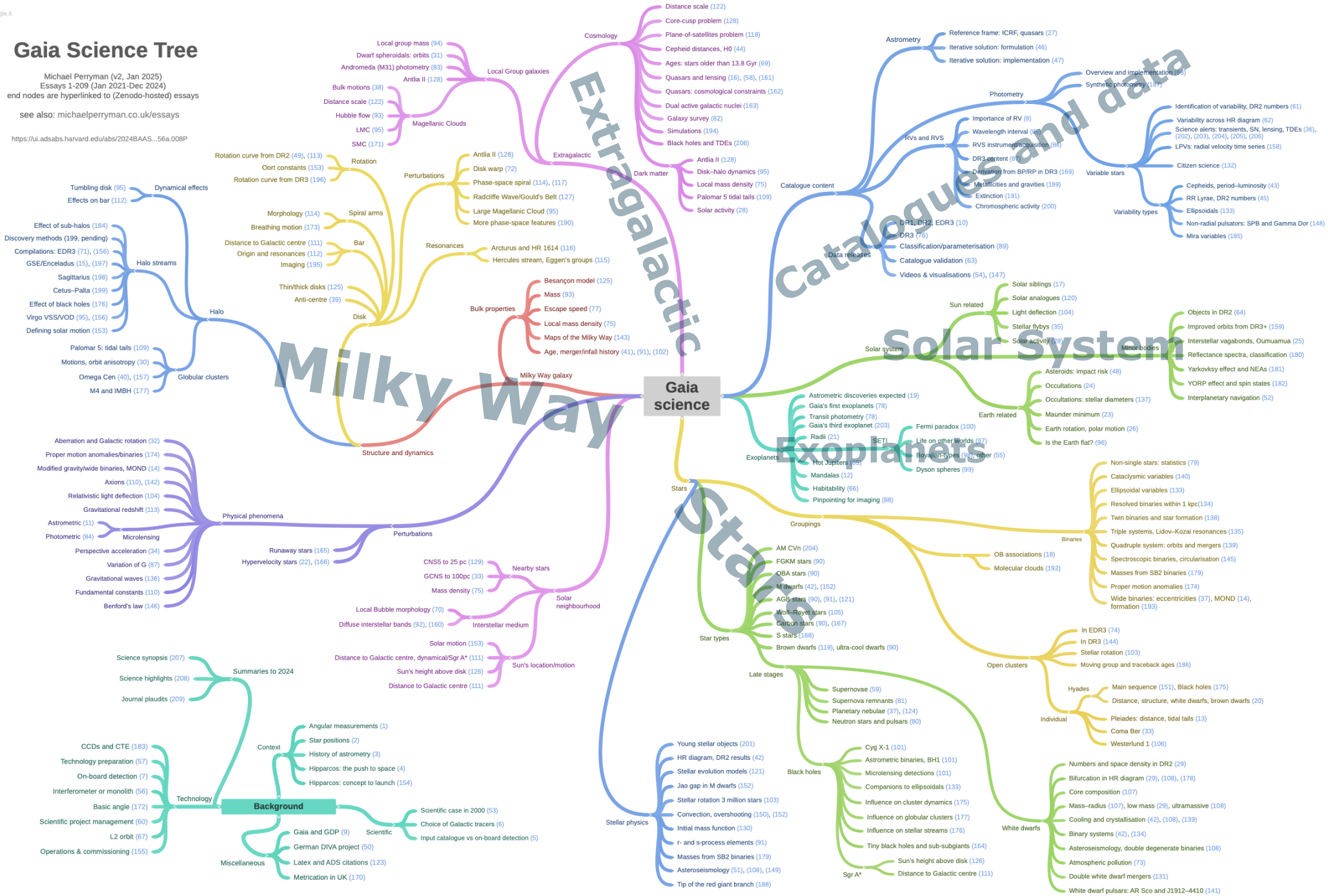
coggle
made for free at coggle.it

Gaia Science Tree

Michael Perryman (v2, Jan 2025)
Essays 1-209 (Jan 2021-Dec 2024)
end nodes are hyperlinked to (Zenodo-hosted) essays

see also: michaelperryman.co.uk/essays

https://ui.adsabs.harvard.edu/abs/2024BAAS...56a.008P



Credit:
Michael Perryman
(source)

SKY-SCANNING COMPLETE FOR ESA'S MILKY WAY MAPPER GAIA

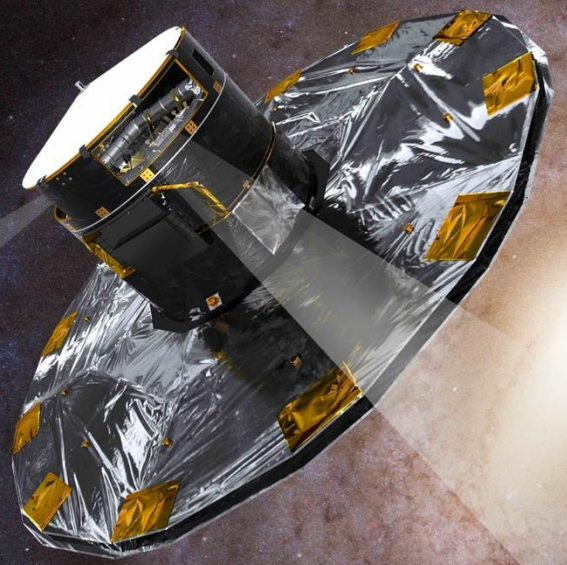
From 24 July 2014 to 15 January 2025, Gaia made more than three trillion observations of two billion stars and other objects, which revolutionised the view of our home galaxy and cosmic neighbourhood.

3 TRILLION
Observations

2 BILLION
Stars & other objects observed

938 MILLION
Camera pixels on board

15 300
Spacecraft 'pirouettes'



580 MILLION
Accesses of Gaia catalogue so far

13 000
Refereed scientific publications so far

2.8 MILLION
Commands sent to spacecraft

142 TB
Downlinked data (compressed)

500 TB
Volume of data release 4
(5.5 years of observations)

55 KG
Cold nitrogen gas consumed

3827
Days in science operations

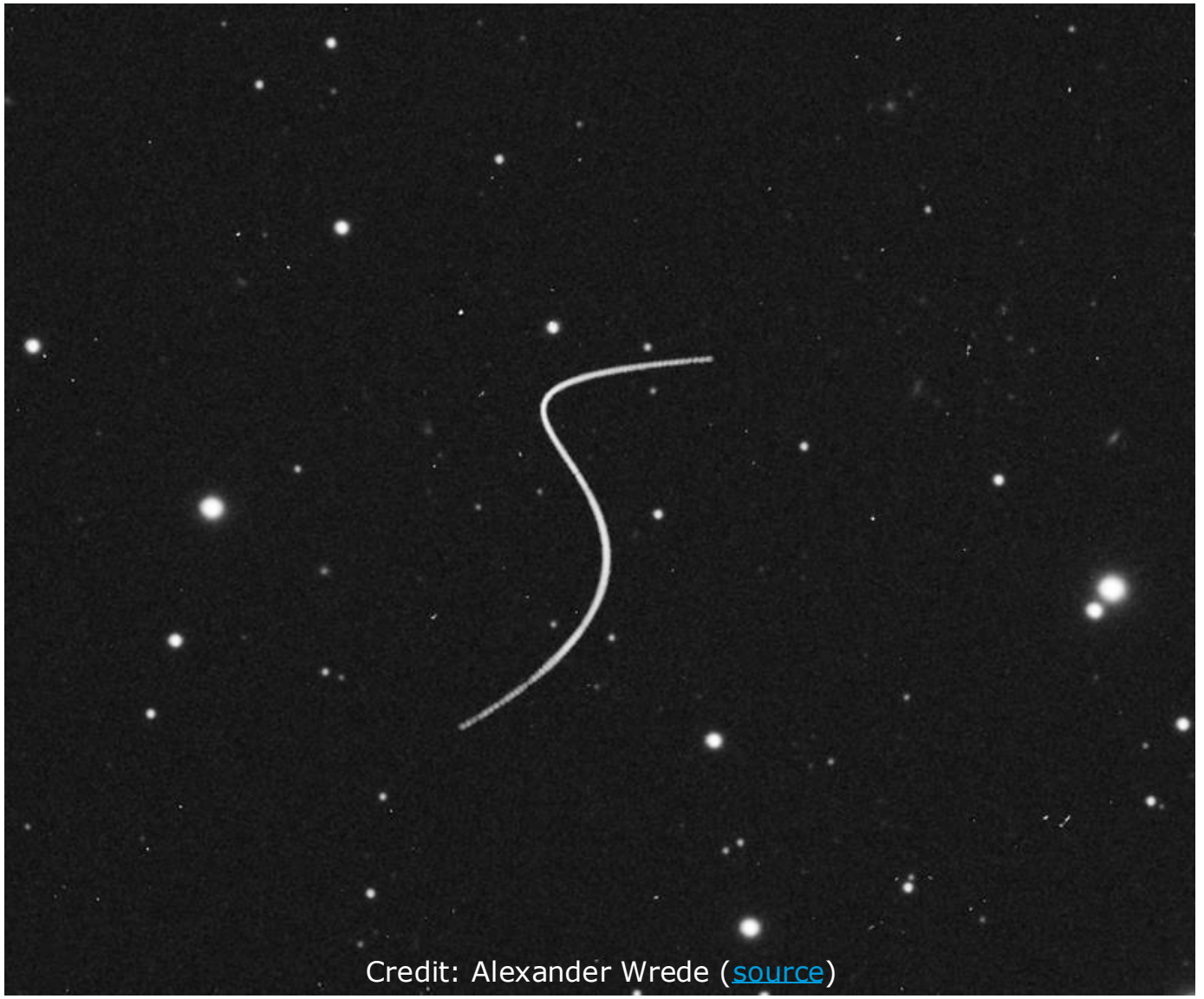
50 000 HOURS
Ground station time used

Credit: ESA/Gaia/DPAC (source)

Gaia spacecraft underwent a series of technical tests

- Gaia technical tests from 15 Jan 2025 – 26 March 2025
  - Special calibrations
  - Spacecraft technology testing
  - Investigation of “basic angle variations”

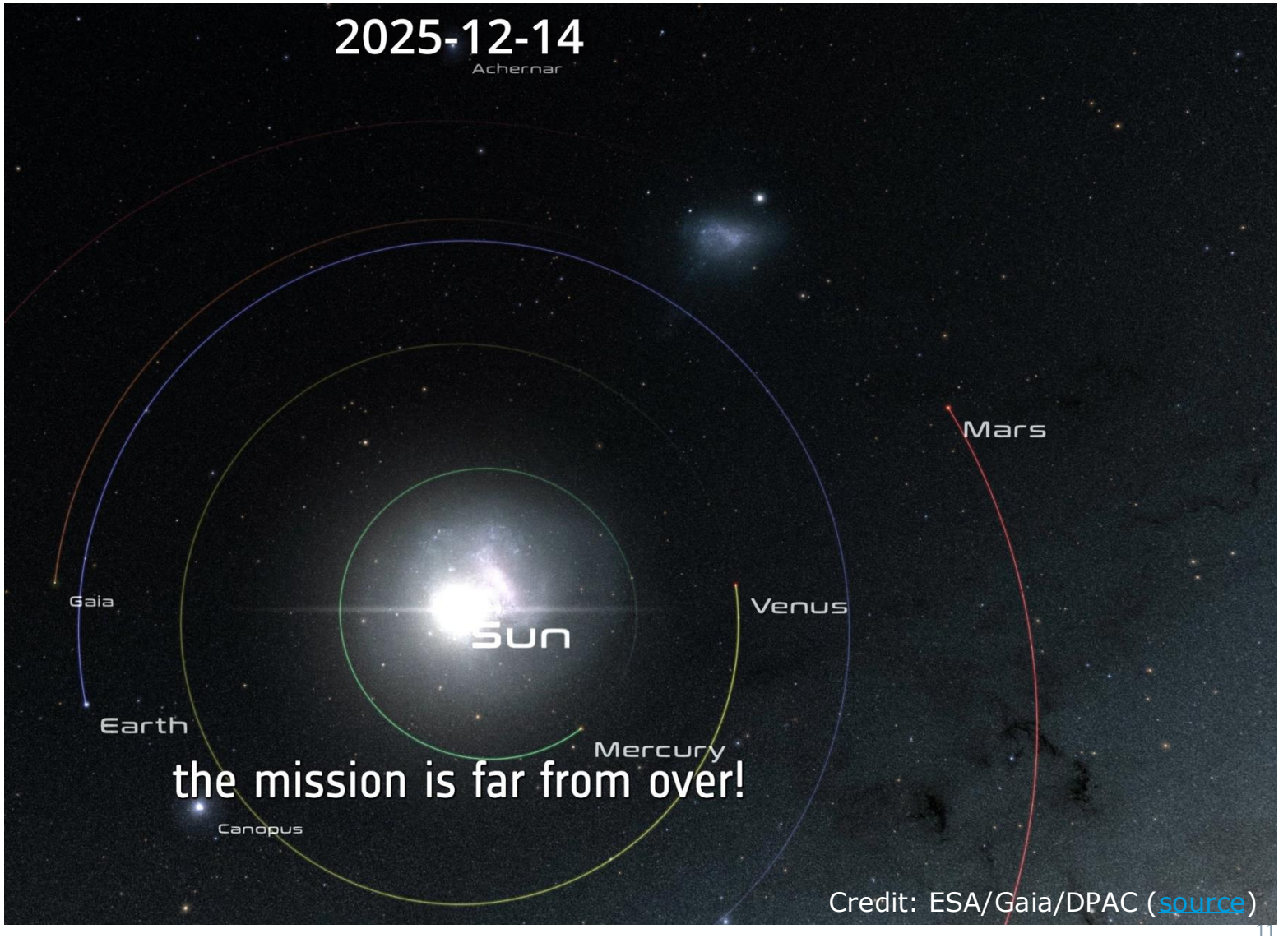
- “Bright” Gaia was observed by citizen astronomers with telescopes worldwide



Credit: Alexander Wrede (source)

Gaia spacecraft was passivated on 27 March 2025

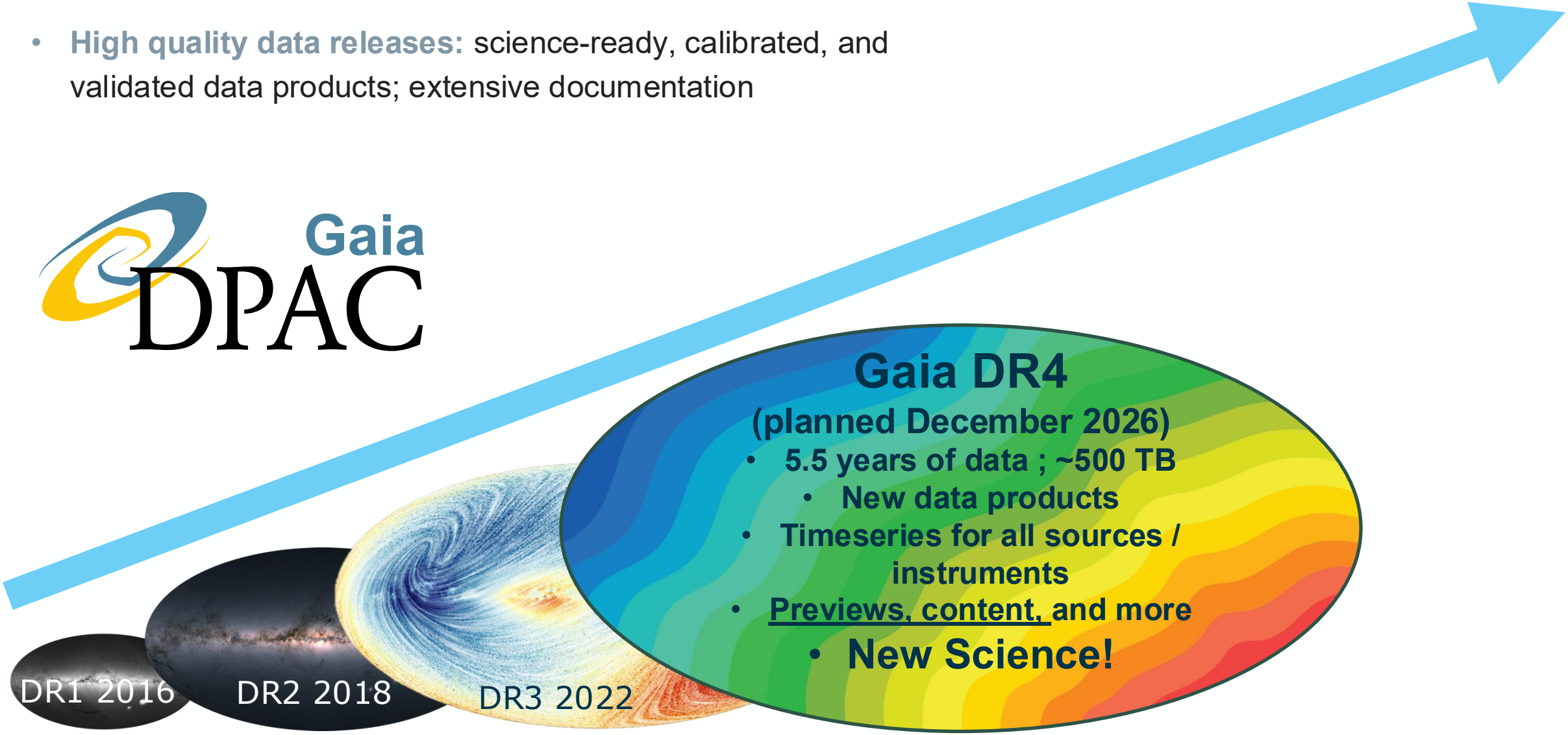
- Gaia exhausted its cold gas and stopped science observations on 15 January 2025
- Gaia was put in a heliocentric orbit and permanently switched off
- ESA news release: https://www.esa.int/Enabling_Support/Operations/Farewell_Gaia!_Spacecraft_operations_come_to_an_end
- Additional information: https://www.cosmos.esa.int/web/gaia/iow_20250327
- This is not the end of the mission



The Gaia collaboration is preparing Gaia Data Release 4

- **High quality data releases:** science-ready, calibrated, and validated data products; extensive documentation

Gaia DPAC



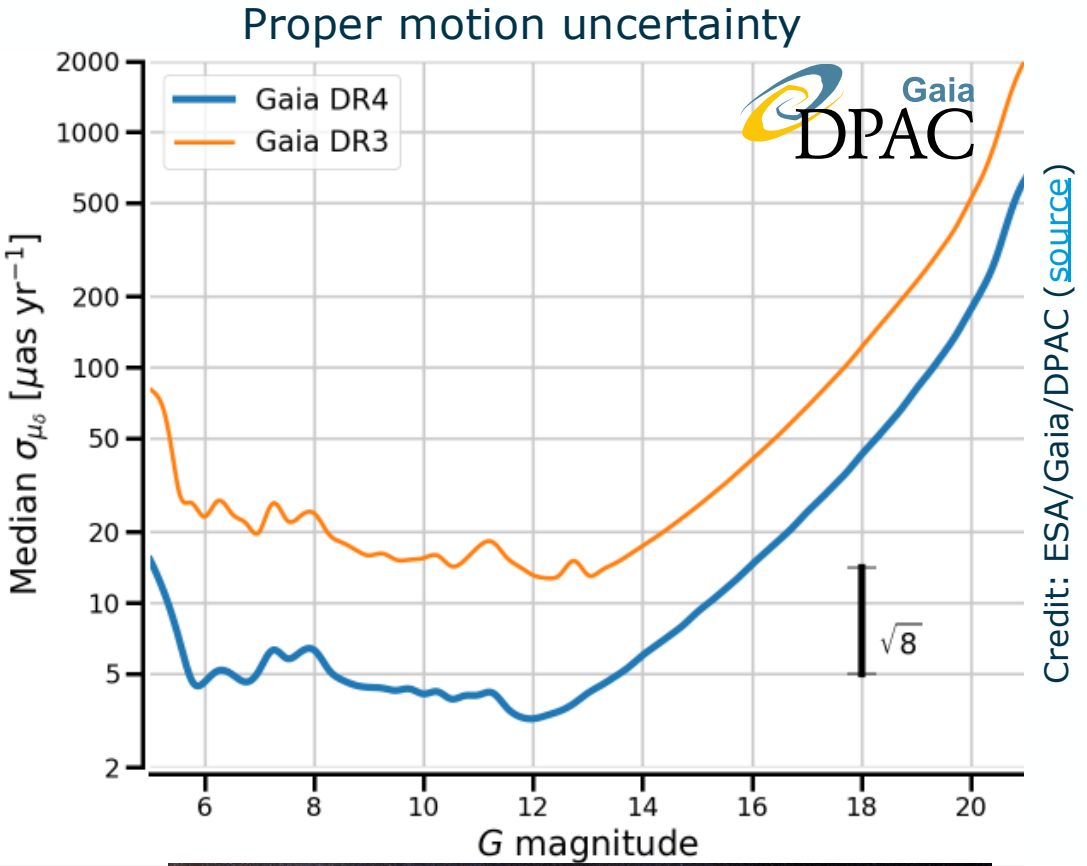
Upcoming Gaia DR4 is the “nominal mission” data release

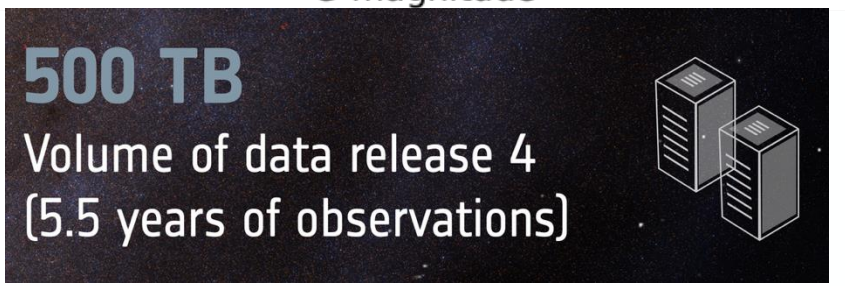
- Gaia DR4 covers 5.5 years (2014-2020) of Gaia data

- More data, longer timespan, better precision, higher accuracy
- >2 billion sources
- More and new data products, including lower-level data:
  - Time series of astrometric data + CCD images
  - Lightcurves: time series of photometric data
  - Radial velocity time series
  - Time series of low- and high-resolution spectra
  - New non-single-source classes, source environment information, crowded field images, and much more

Talks by Brown, Eyer, Ripepi, De Bruijne, Roegiers, and others

- Gaia DR4 will lead to another flurry of discoveries





Overview of Gaia DR4 and its content is public

- Landing page: https://www.cosmos.esa.int/web/gaia/data-release-4
- Content page: https://www.cosmos.esa.int/web/gaia/dr4
- Currently incomplete list of >100 data tables and release items. Updates will be made gradually.

Export

Search: Case Insensitive

Id	Data Model Chapter	Gaia Archive Table Name	Description	Category	New in DR4	Data level	Label
01.01	Main source catalogues	gaia_source	Sources considered to have both high quality astrometry and photometry. In contrast to previous data releases, the parameters in this table are consolidated from several different processing modules.	Main		DL3	General
01.02	Main source catalogues	gaia_source_environment	Astrometry, photometry, and related information for sources detected by the SEAPipe processing in the surrounding of sources in the main Gaia source catalogue.	Main	NEW	DL3	General
01.03	Main source catalogues	all_source_astrometry	Astrometric parameters obtained for all sources (except those in crowded_field_source and gaia_source_environment).	Main	NEW	DL3	Astrometry

Overview of Gaia DR4 and its content is public

Export

Search: NEW Case Insensitive

Id	Data Model Chapter	Gaia Archive Table Name	Description	Category	New in DR4	Data level	Label
13.11	Spectroscopic tables	xp_summary	Auxiliary information about the mean BP/RP spectrum of a given source	Main	NEW	DL3	Photometry
13.09	Spectroscopic tables	xp_epoch_spectrum	Epoch BP and RP spectra collected at transit level for non-SSO sources.	Epoch	NEW	DL2	Spectra
13.10	Spectroscopic tables	xp_epoch_crowding	Information on both treated and untreated crowding for both BP and RP CCDs for a FoV transit	Epoch	NEW	DL2	Photometry
14.19	Variability tables	vari_yso	Variable sources - Young stellar objects variability	Main	NEW	DL4	Variability
14.0N	Variability tables	vari_unsupervised	Unsupervised machine-learning classification for variable sources	Main	NEW	DL4	Classification
14.10	Variability tables	vari_rad_vel_statistics	Statistical parameters of radial velocity time series.	Main	NEW	DL3	Variability

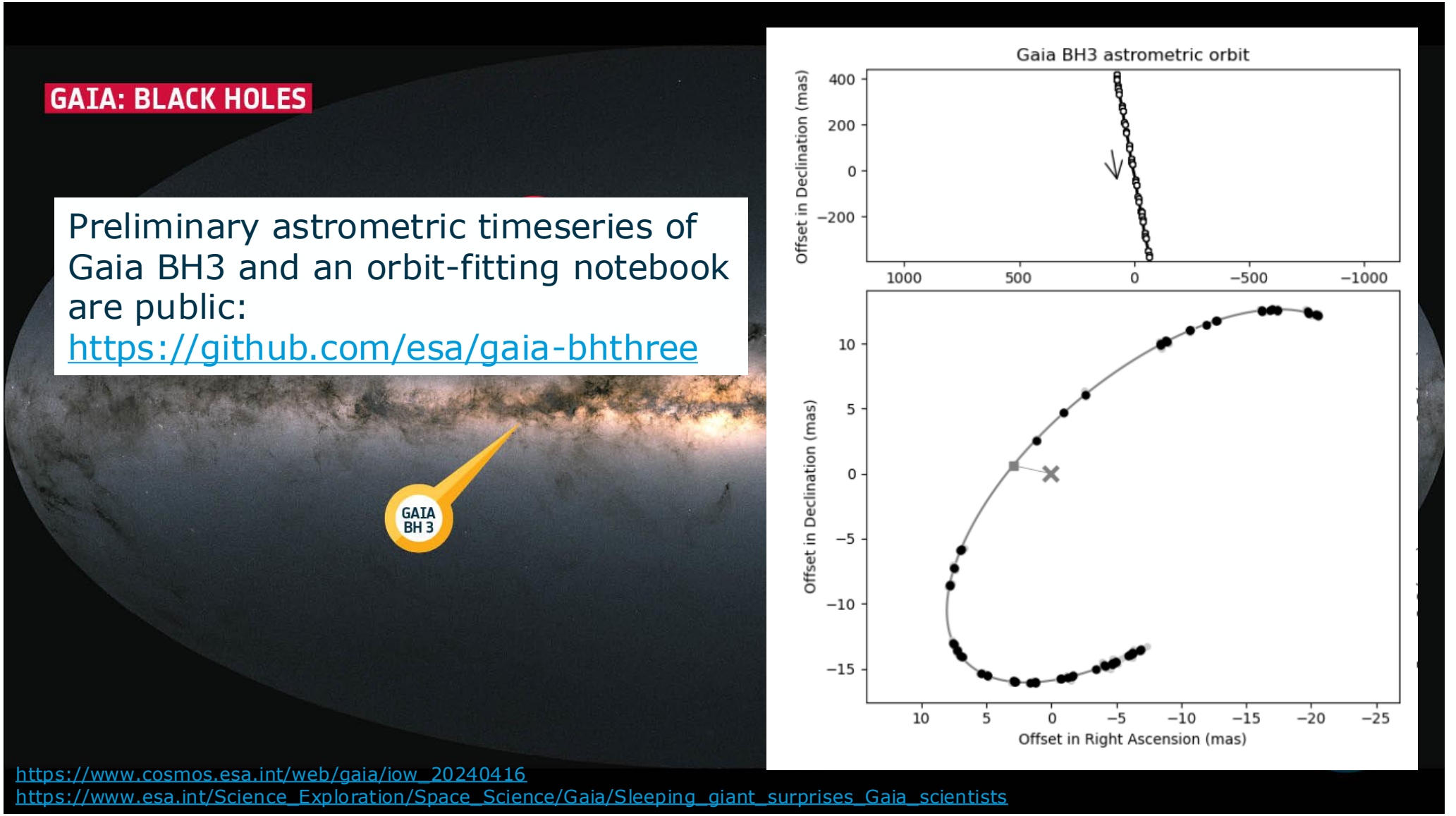
Overview of Gaia DR4 and its content is public

Export

Search: DL2 Aa Case Insensitive

Id	Data Model Chapter	Gaia Archive Table Name	Description	Category	New in DR4	Data level	Label
03.03	Crowded fields	crowded_field_epoch_astrometry	Time series of astrometric data, i.e. the individual astrometric measurements for the sources treated in the special crowded-field processing.	Epoch	NEW	DL2	Astrometry
03.0N	Crowded fields	crowded_field_epoch_photometry	Time series of photometric data, i.e. the individual photometric measurements for the sources treated in the special crowded-field processing.	Epoch	NEW	DL2	Photometry
06.06	Extra-galactic tables	grav_lens_observation	Observations used for gravitational lens search.	Gravitational lensing		DL2	Classification
08.01	Epoch-level astrometric, photometric and image products	epoch_astrometry	Time series of astrometric data, i.e. individual astrometric measurements for all sources treated in the main astrometric processing.	Epoch	NEW	DL2	Astrometry

Gaia DR4 preview: astrometric timeseries of Gaia BH3

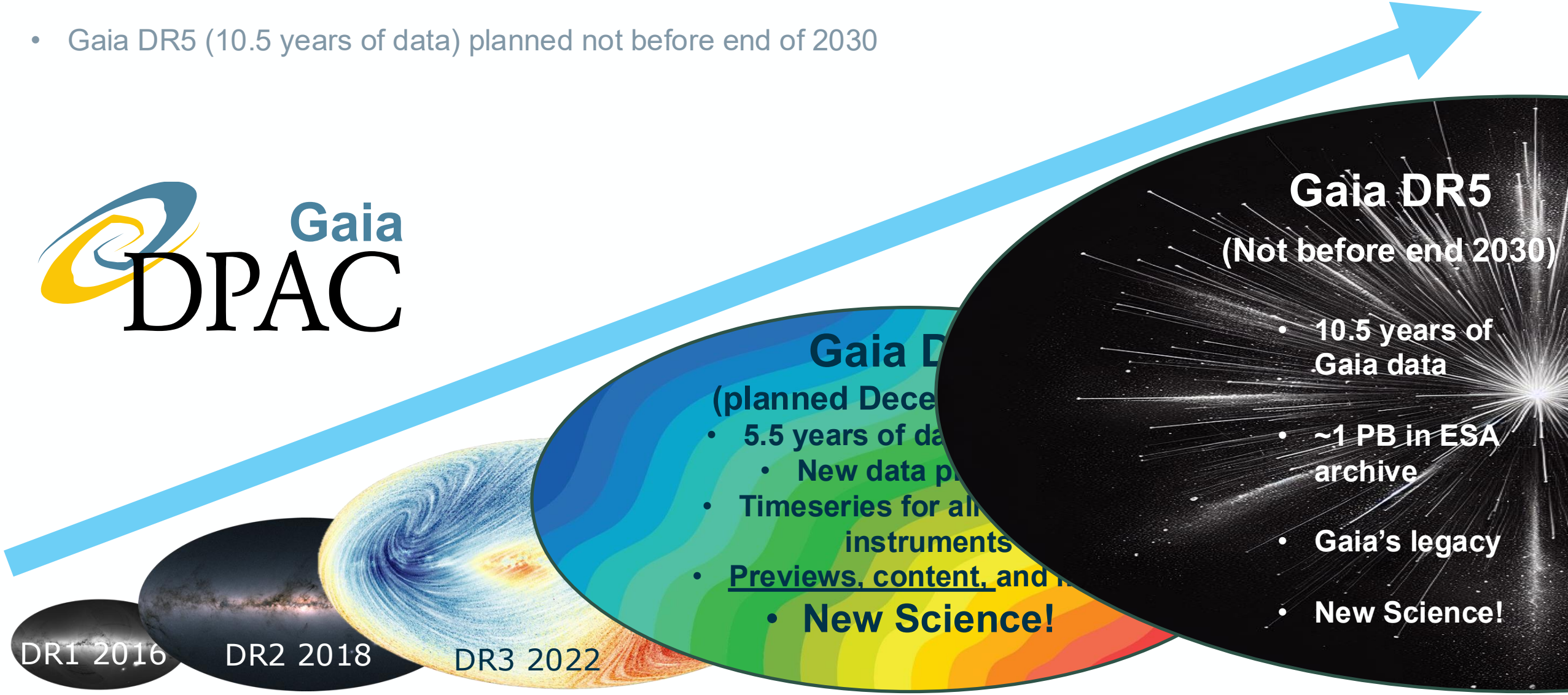


Preliminary astrometric timeseries of Gaia BH3 and an orbit-fitting notebook are public:
https://github.com/esa/gaia-bhthree

https://www.cosmos.esa.int/web/gaia/iow_20240416
https://www.esa.int/Science_Exploration/Space_Science/Gaia/Sleeping_giant_surprises_Gaia_scientists

Gaia Data Release 5 will constitute the mission's legacy

- Gaia DR5 (10.5 years of data) planned not before end of 2030



Gaia and ESA resources at your fingertips

- Please acknowledge the use of Gaia data in your research: https://www.cosmos.esa.int/web/gaia-users/credits
- Information for Gaia scientific community: https://www.cosmos.esa.int/web/gaia

Talk by Roegiers

- Receive Gaia data user notifications by signing up for the Gaia Bulletin emailing list: https://www.cosmos.esa.int/web/gaia/bulletin
- Interested in boosting the visibility of your Gaia-related research with an ESA news release? See https://www.cosmos.esa.int/web/gaia/communicating-your-results
- Information on **ESA Archival Research Visitor Programme**: https://www.cosmos.esa.int/web/esdc/visitor-programme (next deadline 31 October 2025)
- Information on **ESA Research Fellowships in Space Science**: https://www.cosmos.esa.int/web/space-science-faculty/opportunities/research-fellowships (next call end August with a deadline mid-September 2025)

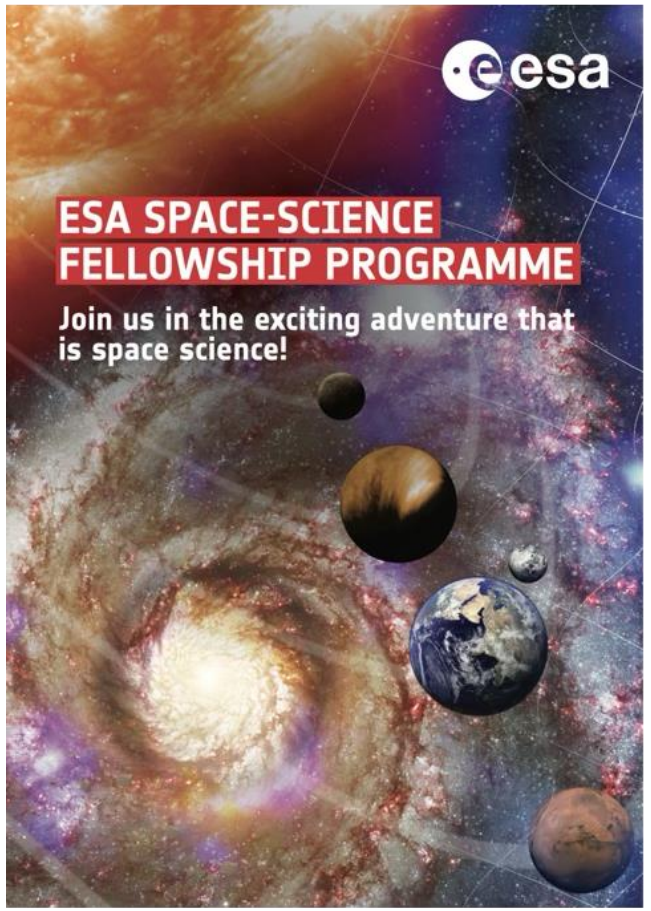
Conclusions

- Gaia delivered on the promise to revolutionise our understanding of the Milky Way.
- Gaia became a new backbone of modern astronomy and astrophysics.
- We wave goodbye to the Gaia spacecraft, but the Gaia mission continues strong.
- Gaia Data Release 4 is planned for December 2026.
- Gaia Data Releases 4 (and 5) will unleash the full scientific power of the Gaia survey.
- The best from Gaia is ahead of us.



Credit: ESA (source)

ESA Science Research Fellowship



What?

- **independent postdoctoral fellowship for ESA State nationals**
- research project covering any topic in space science
- 2 + 1 years (proposal for 3rd year extension)

Where?

- ESTEC (Netherlands), ESAC (Spain) or STScI (USA)

Why?

- **100% research time** (optionally <20% functional work, e.g. archive/data science, citizen science, operations, calibration, communication)
- insights into ESA environment & activities
- mentoring from senior ESA Science Faculty members
- training available (e.g. spacecraft design, soft skills, management)
- 3000-4600€ net monthly salary (depending on location & experience)
- comprehensive health coverage

Website: https://www.cosmos.esa.int/fellowship

contact: victoria.grinberg@esa.int